Renewable Electricity Trends

Nishant Jha

Independent Analysis

Energy Data Project

Project Report

October 14, 2026

Renewable Electricity Trends

Overview

This report examines renewable electricity in India, Germany, and Brazil from 2000 to 2024. It uses the Our World in Data Energy dataset and a focused Python workflow to compare renewable shares, generation, and source mix. The notebook and local data file make the analysis reproducible (Danchev, 2022).

Dataset Description

The September 10, 2026 release of the Our World in Data Energy dataset contains 19,494 rows and 130 columns, with country or region observations from 1700 through 2025. It combines electricity and energy measures from the Energy Institute, Ember, the United States Energy Information Administration, and historical reconstructions (Our World in Data, 2026).

The analysis focuses on three countries and 25 annual observations per country, from 2000 through 2024. The key measures are renewable electricity share, renewable generation in terawatt hours, and hydro, wind, solar, bioenergy, and other renewable shares. The comparison ends in 2024 by design so each country uses the same fixed period; 2025 is available in the source but is outside this scope.

Workflow Description

The notebook loads the complete CSV with Pandas and displays its first rows. It standardizes column names, selects the three countries and fixed year range, converts analysis measures to numeric values, removes duplicate country and year keys, and excludes records without the renewable share measure. It then summarizes start and end values, reviews country level trends, and produces three labeled charts for share, generation, and source mix.

A recorded Python environment and a local copy of the dated source file support repeatable runs. A notebook based workflow also keeps code, outputs, and explanations together, which supports reproducible data science practice (Danchev, 2022).

Key Decisions and Assumptions

The analysis uses percentage share to compare the proportion of electricity from renewable sources and terawatt hours to show the scale of generation. The EDA compares the first and last years and checks the highest share, while retaining every year for charting. This avoids treating a country with more total electricity generation as automatically having a larger renewable share.

Figure 1 was designed to show the percentage trend over time. Figure 2 shows renewable generation volume over the same period. Figure 3 compares source composition in 2000 and 2024. These views are useful because renewable electricity transitions differ across countries and sources; a peer reviewed G20 study also reports substantial country variation and growth in wind and solar (Sokulski et al., 2022).

Results and Interpretation

Germany's renewable share rose from 6.22% in 2000 to 58.64% in 2024, an increase of 52.42 percentage points. India's share rose from 14.05% to 19.81%, while Brazil had the highest share in both endpoint years and moved from 89.49% to 87.33%.

Renewable generation increased in all three countries. Brazil rose from 312.26 to 651.26 terawatt hours, India from 80.27 to 403.39, and Germany from 35.41 to 290.86. The source mix chart shows hydropower as the largest renewable source in Brazil and India in 2024. Germany's renewable mix is led by wind, with solar also contributing visibly. These results describe the selected series and do not establish the causes of change.

Responsible Practice (Bias and Data Quality)

Dropping missing years could bias a comparison if countries with weaker historical reporting were excluded more often. Filling missing values could make a smooth but unsupported trend, and failing to remove duplicate country and year rows could give some years extra weight. The selected 75 country year records have complete values for the measures used, and no missing value was imputed. The code checks duplicates and only drops records missing the main renewable share measure.

The dataset combines providers and historical reconstructions, so methods and revisions may differ over time. The analysis assumes the published country definitions and measures are comparable within this release. A later review should compare source notes across indicators and repeat the analysis with generation and share definitions from one provider when consistent coverage is available.

Reproducibility

A reader can install the recorded packages from requirements.txt, open data_workflow.ipynb, and run the cells from top to bottom with data/owid_energy.csv in place. The full source file and its catalog link identify the dataset release. Multiple commits document development, and a separate development branch provides a place to test changes before they reach the default branch.

References

Danchev, V. (2022). Reproducible data science with Python: An open learning resource. Journal of Open Source Education, 5(56), Article 156. https://doi.org/10.21105/jose.00156

Our World in Data. (2026, September 10). Energy dataset [Data set]. https://catalog.ourworldindata.org/energy/owid_energy/

Sokulski, C. C., Barros, M. V., Salvador, R., Broday, E. E., & de Francisco, A. C. (2022). Trends in renewable electricity generation in the G20 countries: An analysis of the 1990 to 2020 period. Sustainability, 14(4), Article 2084. https://doi.org/10.3390/su14042084